

Pseudoword Generator — PTAF & Orthographic Stats Matched

Generates pseudowords for psycholinguistic experiments. For each real English word in an input CSV, the script produces a pronounceable, word-like pseudoword whose Phonological Total All-neighbors Frequency (PTAF) is matched to the real word's target PTAF, while dynamically calculating orthographic neighborhood statistics (OTAN, OTAF) and reporting difference metrics.

Requirements

- Python 3.10+
- CLEARPOND English database (offline download)
- Wuggy pseudoword generator
- Rich terminal rendering library

Setup

```
python3 -m venv .venv
source .venv/bin/activate
pip install wuggy rich
```

The first run automatically downloads the Wuggy orthographic_english language plugin (~1 MB) from the Wuggy repository.

Input files

Your word list (--input)

A CSV with at minimum the following columns (extra columns are ignored, and statistics rows like MEAN or T-test are automatically skipped):

Column	Description
Word	Real English word (lowercase)
Length (Ortho)	Orthographic letter length of the word
Word_PTAN	Phonological Total All-neighbors Size (PTAN from CLEARPOND)
Word_PTAF	Phonological Total All-neighbors Frequency (ePTAF mean from CLEARPOND)
Word_OTAN	Orthographic Total All-neighbors Size (OTAN from CLEARPOND)
Word_OTAF	Orthographic Total All-neighbors Frequency (eOTAF mean from CLEARPOND)

Example (data/words.csv):

```
Word,Length (Ortho),Word_PTAN,Word_PTAF,Word_OTAN,Word_OTAF
chair,5,37,220.2279,7,27.6163
stone,5,18,6.9662,12,15.1356
brick,5,20,42.6882,9,11.2505
```

CLEARPOND English database (--clearpond)

Download the offline English database from clearpond.northwestern.edu. The expected file is englishCPdatabase2.txt (tab-delimited, ~28 k words). The companion header file clearpondHeaders_EN.txt must sit in the same directory as the database file.

Place both files under data/englishCPdatabase2/:

```
data/
├── englishCPdatabase2/
│   ├── englishCPdatabase2.txt
│   └── clearpondHeaders_EN.txt
```

Usage

The script reads the clean input CSV file and writes its output to a separate output CSV file, leaving the input file untouched:

```
.venv/bin/python pseudowords.py \
  --input data/words.csv \
  --clearpond data/englishCPdatabase2/englishCPdatabase2.txt \
  --output data/pseudowords_output.csv \
  --tolerance 0.20 \
  --candidates 1000 \
  --seed 23
```

Options

Flag	Default	Description
--input	(required)	Clean input CSV with real words and statistics
--clearpond	(required)	Path to englishCPdatabase2.txt
--output	pseudowords_output.csv	Output CSV path (separate from the input file, e.g. data/pseudowords_output.csv)
--tolerance	0.20	Maximum allowed relative PTAF difference (0.20 = 20%)
--candidates	1000	Number of Wuggy pseudoword candidates evaluated per word
--seed	23	Random seed (affects fallback generator only)

Output Columns

The output CSV contains a comprehensive set of matched results and statistics:

Column	Type	Description
Word	str	Original real word
Length (Ortho)	int	Real word orthographic length
Word_PTAN	int	Real word phonological neighborhood size
Word_PTAF	float	Real word precalculated target PTAF mean
Word_OTAN	int	Real word orthographic neighborhood size
Word_OTAF	float	Real word precalculated target OTAF mean
Pseudoword	str	Generated pseudoword (orthographic, readable)

Column	Type	Description
Consonant_Pseudoword	str	Random consonant-only pseudoword of the same length
Pseudoword_PTAN	int	Computed phonological neighborhood size of the pseudoword
Pseudoword_PTAF	float	Computed phonological neighborhood frequency mean of the pseudoword (same convention as CLEARPOND's ePTAF)
Pseudoword_OTAN	int	Computed Coltheart's distance-1 orthographic neighborhood size of the pseudoword
Pseudoword_OTAF	float	Computed orthographic neighborhood frequency mean of the pseudoword
PTAF_RelDiff_Pct	float	Relative difference in percent between the pseudoword and real word PTAF means (the matching criterion)
OTAN_Diff	int	Direct integer difference in orthographic neighborhood size: Pseudoword_OTAN - Word_OTAN
OTAF_RelDiff_Pct	float	Relative difference in percent between the pseudoword OTAF mean and the real word OTAF mean
Status	str	MATCHED (within tolerance), BEST_AVAILABLE (closest match found, exceeds tolerance), or NO_TRANSCRIPTION (existing pseudoword whose onset/rime is not attested in CLEARPOND, so no phonological statistics could be computed)
Method	str	existing (pseudoword taken from the input), wuggy (Wuggy-generated), or fallback (phoneme-mutation fallback)

How it works

1. Load CLEARPOND

The CLEARPOND English database is loaded into memory, extracting the word's orthographic, phonotactic, and neighborhood frequency records. A reverse index `phoneme_tuple → [(word, freq)]` is built for fast neighbor lookups.

2. DISC phoneme format

CLEARPOND uses a dot-separated DISC notation where each token is a single phoneme. Multi-character tokens represent single phonemes:

Token	Phoneme	Example
tS	/tʃ/	ch in chair
dZ	/dʒ/	j in jump
oU	/oʊ/	o in stone
eI	/eɪ/	a in cable
r0	/r/	r in brick

Token	Phoneme	Example
36	/ɜ:/	ur in journal

3. Build onset/rime phoneme maps

For every CLEARPOND word, the script extracts its orthographic onset (letters before the first vowel) and rime (first vowel onward) to build onset/rime phone mappings:

```
onset_map["ch"] → ('tʃ',)
rime_map["arp"] → ('Ar', 'p')
```

These are used to approximate the phoneme sequence of any Wuggy pseudoword. If a pseudoword's onset or rime is not attested in CLEARPOND, no transcription is attempted: generated candidates are skipped, and existing pseudowords from the input are reported with Status = NO_TRANSCRIPTION. (Guessing — e.g. reusing the source word's onset/rime — could reconstruct the source word's own phonology and silently score the real word instead of the pseudoword.)

4. Generate candidates with Wuggy

Wuggy generates candidate pseudowords using subsyllabic substitution (onset, nucleus, coda), guaranteeing English-like phonotactics and non-existence in the English lexicon.

Fallback: If Wuggy cannot handle a word, or none of its candidates has a transcribable onset/rime, the script falls back to mutating 1–3 phonemes of a same-length real-word template while preserving consonant/vowel category at each mutated position.

5. Scoring & Matching

- PTAF Matching: For each candidate, the script enumerates all phonological neighbors at edit distance 1 (one-phoneme substitution, deletion, or insertion) and computes Pseudoword_PTAF as the mean frequency of those neighbors (frequency sum ÷ neighbor count) — the same convention as CLEARPOND's precalculated ePTAF, so the pseudoword statistic is directly comparable to the Word_PTAF target. The candidate whose mean lies closest to the target wins, and is accepted as MATCHED if the relative difference is within --tolerance.
- PTAF Output Values: Pseudoword_PTAF writes the neighbor-frequency mean, and PTAF_RelDiff_Pct is the relative difference between this mean and the target mean.

6. Orthographic Neighborhood Calculations

For the selected best pseudoword, orthographic statistics are computed dynamically using Coltheart's distance-1 definition:

- Orthographic Neighborhood Size (Pseudoword_OTAN): Count of unique real CLEARPOND words obtained by exactly one single-letter deletion, substitution, or insertion.
- Orthographic Neighborhood Frequency (Pseudoword_OTAF): The average (mean) frequency of those orthographic neighbors, computed as OTAF_sum / OTAN to remain mathematically consistent with the database precalculated mean Word_OTAF.
- OTAN Difference (OTAN_Diff): The direct integer difference Pseudoword_OTAN - Word_OTAN.
- OTAF Relative Difference (OTAF_RelDiff_Pct): Relative difference in percent between the pseudoword and real word's orthographic mean frequencies.

Notes

- Static Type Safety: Fully type-safe and validated using the pyrefly static type analysis.
- Safety: The clean input file data/words.csv remains untouched, and all generation outputs are piped to your designated output file (e.g. data/pseudowords_output.csv).